

INC-2026-0812 - Cart Token Validation Failures After Checkout Release 4.12.0

Incident Report | Engineering

Document ID	INC-2026-0812	Seller ID	S001
Department	Engineering	Classification	CONFIDENTIAL
Owner	Vikram Singh	Created Date	2026-08-14
Related Scenario ID	SC-010	Document Type	Incident Report

1. Incident Summary

Field	Value
Incident ID	INC-2026-0812
Severity	SEV-2
Service	Checkout Service
Release	4.12.0
Detected	2026-08-12 11:44 IST
Mitigated	2026-08-12 12:36 IST
Resolved	2026-08-13 16:00 IST
Incident owner	Vikram Singh, Engineering
Seller reporting	S001 - Aurelia Home Decor
Related ticket	TKT-2026-0812-308

Checkout Service release 4.12.0 changed the cart token cache key format while continuing to sign tokens using the previous key namespace. During the rolling deployment, instances running the two versions produced and validated tokens inconsistently. Approximately 7 percent of cart-to-payment transitions failed with an invalid cart token error over a 116-minute window.

2. Impact

Dimension	Detail
Failed checkouts	1,470 platform-wide
Failure rate at peak	7.2 percent of cart-to-payment transitions
Duration	116 minutes, 10:40 to 12:36 IST
Sellers affected	All sellers; S001 most visibly
S001 failed checkouts	312
S001 recovered same day	244
Estimated net lost orders, S001	68
Data loss	None; carts were preserved

The impact was platform-wide but was reported by a single seller. Aurelia was running a pre-festive campaign that day, so its traffic was elevated and its customer service team noticed the pattern within an hour. Other sellers experienced the same failure rate without recognising it.

3. Detection

Detected by a seller through Support, not by monitoring.

Aurelia’s seller operations lead called Support at 11:26 IST after eleven buyer complaints in an hour. Support reproduced the failure at 11:38, confirmed it was not seller-specific by reproducing it against a different seller’s listing, and escalated at 11:44.

Platform monitoring did register the change: checkout success rate had dropped from 98.9 percent to 91.7 percent at 10:40. No alert fired because the success-rate alert threshold was set at 85 percent, chosen years earlier to catch outages rather than partial degradations.

Time from occurrence to detection: 64 minutes.

4. Timeline

Time (IST)	Event
12 Aug 10:31	Rolling deployment of 4.12.0 begins, 3 of 12 instances
12 Aug 10:40	First token validation failures observed
12 Aug 10:52	Deployment proceeds to 6 of 12 instances; failure rate rises
12 Aug 11:14	Deployment reaches 9 of 12 instances; failure rate peaks at 7.2 percent
12 Aug 11:26	Seller calls Support
12 Aug 11:38	Support reproduces; confirms not seller-specific
12 Aug 11:44	Escalated to Engineering; incident opened
12 Aug 11:51	Correlated with 4.12.0 rollout start time
12 Aug 11:58	Deployment halted
12 Aug 12:02	Rollback to 4.11.7 initiated per Deployment Rollback Procedure
12 Aug 12:36	Rollback complete; failure rate returns to baseline
12 Aug 13:15	Support confirms successful checkouts on affected listings
12 Aug 17:00	Root cause confirmed in code review
13 Aug 14:20	4.12.1 built with version compatibility guard
13 Aug 16:00	4.12.1 deployed successfully

5. Symptoms

- Buyers received "invalid cart token, please refresh" at the payment step.
- Failures were intermittent for the same buyer: a retry often succeeded.
- Failure rate tracked the proportion of instances running the new version.
- No errors appeared in the Checkout Service error logs, because a failed token validation was handled as an expected business outcome rather than as an exception.
- Cart contents were never lost; only the transition to payment failed.

The intermittency was the diagnostic key. A deterministic failure would have suggested a logic defect. A failure that depended on which instance served the request suggested a version mismatch.

6. Affected Services

Service	Role	State
Checkout Service	Issued and validated cart tokens	Defective in 4.12.0
Token cache	Stored tokens under the new key format	Functioning as designed
Signing key store	Provided namespaced signing keys	Functioning as designed
Payment Service	Rejected requests without a valid token	Functioning as designed
Load balancer	Distributed requests across mixed versions	Functioning as designed

7. Investigation

Release 4.12.0 introduced a cache key change intended to support multi-region cart replication. The key gained a region prefix:

- 4.11.7: `cart:{cart_id}`
- 4.12.0: `cart:{region}:{cart_id}`

Tokens are signed with a key resolved from a namespace derived from the cache key. The change updated the cache key construction but did not update the namespace derivation, which continued to use the unprefixed form.

In a homogeneous deployment this would have been harmless: every instance would write and read consistently. The failure required a mixed fleet.

Issuing instance	Validating instance	Outcome
4.11.7	4.11.7	Success
4.12.0	4.12.0	Success
4.11.7	4.12.0	Failure: token not found under prefixed key
4.12.0	4.11.7	Failure: namespace mismatch on signature check

With nine of twelve instances upgraded, the probability of a cross-version pair on any given checkout was approximately 37 percent, and roughly one in five of those produced a user-visible failure after internal retries. This matches the observed 7.2 percent peak.

Testing had not caught it because the staging environment deploys all instances simultaneously. There is no mixed-version state in staging, so the only failing combinations never occurred.

8. Root Cause

Release 4.12.0 changed the cart token cache key format to include a region prefix but did not change the corresponding signing key namespace derivation. During the rolling deployment, instances on 4.11.7 and 4.12.0 issued and validated tokens under incompatible key schemes, causing validation failures for any checkout whose requests crossed versions.

9. Contributing Factors

- 1. **No mixed-version testing.** Staging deploys atomically and cannot reproduce a rolling deployment state.
- 2. **No compatibility guard.** Nothing in the release prevented a token format change from being deployed without a transition period.
- 3. **Alert threshold tuned for outages.** A drop from 98.9 to 91.7 percent success was invisible to an alert set at 85 percent.
- 4. **Token validation failures were not logged as errors.** The service treated them as normal business outcomes, so error dashboards were clean throughout.
- 5. **The change description understated the risk.** The release note described the change as adding multi-region cache support. It did not identify a change to a cross-instance data contract.

10. Mitigation

The deployment was halted at 11:58 and rolled back to 4.11.7 starting at 12:02, following the Deployment Rollback Procedure. Rollback completed at 12:36. Failure rate returned to baseline within four minutes of the last instance reverting.

Rollback was chosen over rolling forward because the fix was not yet understood at 12:02 and the failure mode was known to disappear once the fleet was homogeneous.

Time from escalation to mitigation: 52 minutes.

11. Resolution

Release 4.12.1 was built with three changes:

- 1. Signing key namespace derivation updated to match the new cache key format.
- 2. A dual-read transition: 4.12.1 validates tokens under both the prefixed and unprefixed schemes, so a mixed fleet is tolerated during any future rolling deployment of this change.
- 3. An explicit token format version embedded in the token itself, so a validating instance can determine which scheme applies rather than inferring it.

4.12.1 was deployed on 13 August at 16:00 with no failures observed.

12. Validation

Check	Method	Result
Mixed-version tolerance	Deployed 4.12.1 alongside 4.11.7 in staging	No validation failures across 40,000 synthetic checkouts
Failure rate after release	Checkout success rate monitored for 72 hours	98.9 to 99.1 percent, at baseline
Recovered carts	Carts from the incident window checked for integrity	All intact; no cart data lost
Seller confirmation	Support verified checkouts on affected listings	Confirmed working 12 Aug 13:15

13. Preventive Actions

ID	Action	Owner	Target	Status
PA-1	Dual-read transition for token format changes	Vikram Singh	2026-08-13	Complete
PA-2	Explicit token format version field	Vikram Singh	2026-08-13	Complete
PA-3	Mixed-version stage in the deployment pipeline	Engineering	2026-10-09	In progress
PA-4	Checkout success-rate alert lowered to 96 percent	Engineering	2026-08-18	Complete
PA-5	Token validation failures logged as errors with a rate metric	Vikram Singh	2026-08-25	Complete
PA-6	Release notes to declare cross-instance contract changes	Engineering	2026-09-30	Complete

PA-3 is the substantive one. Until a mixed-version stage exists in the pipeline, this class of defect can only be caught by review discipline, which failed here.

14. Follow-Up

A seller notification capability was requested through Support: Aurelia's seller operations lead asked to be told when a platform incident is in progress rather than learning of it from her own customers. This is with Operations for scoping and is outside the scope of this incident.

Post-incident review held 14 August 2026. The review noted that the incident was detected by a seller despite being visible in platform metrics for 64 minutes, and that the alert threshold, not the defect, was the more important finding.